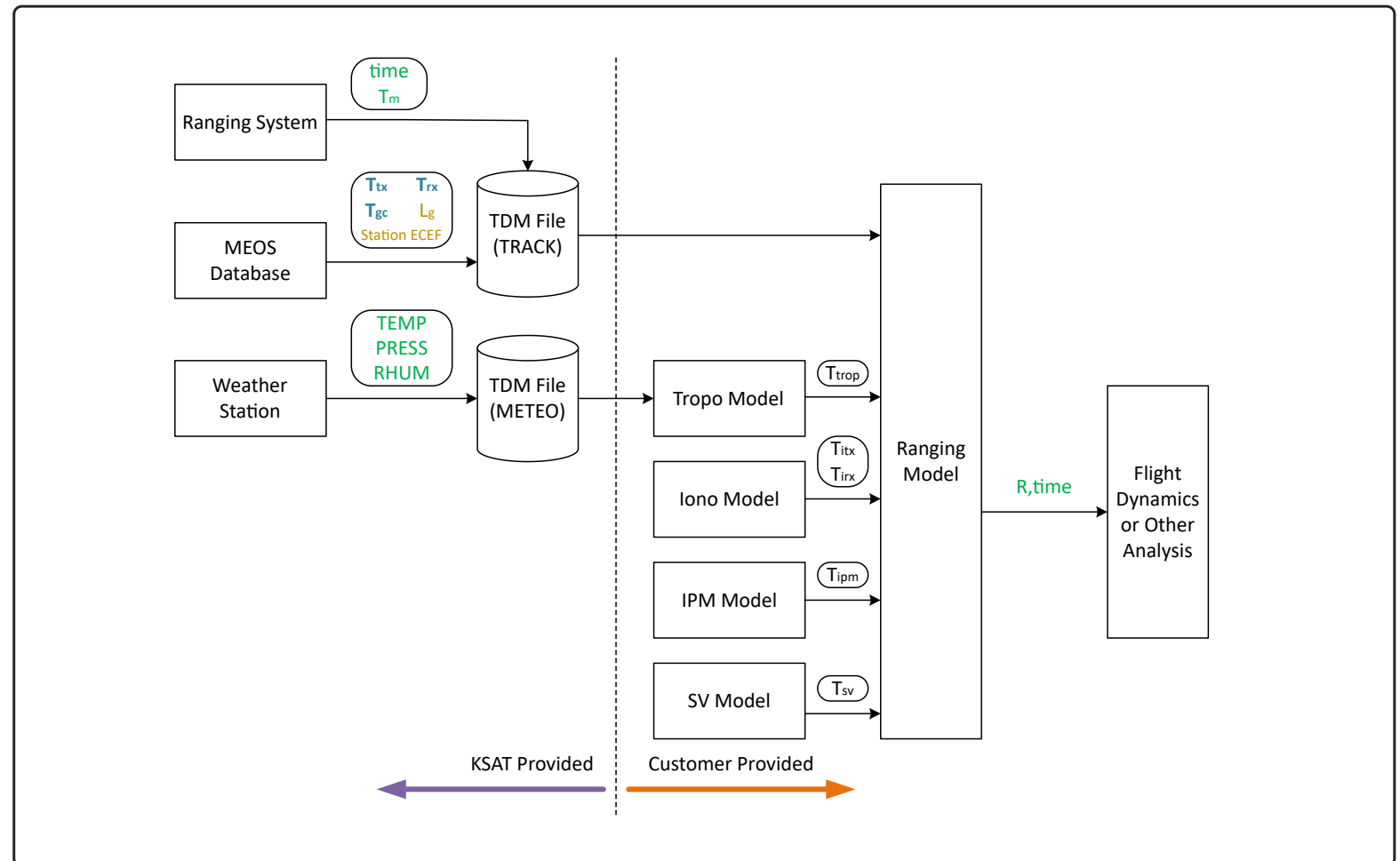


Ranging Model

This page provides an overview of how ranging measurements should be processed.

The ranging model data flow is shown in the following figure.



This data flow model includes the demarcation between KSAT-supplied data and customer-supplied data and models.

KSAT provides raw measurements from the modem, certain constants pertaining to the remote site, and local weather data correlated with the range measurements. These items are provided to the customer in TDM files.

The customer is responsible for providing a tropospheric model (which may or may not use the KSAT-provided tropo data), an ionospheric model, an interplanetary medium model, and a spacecraft model. The customer also must implement the ranging model described [on this page](#) to produce the time-tagged range values prior to processing by flight dynamics or other required analysis.

A more detailed explanation of the delay model used for this ranging algorithm can be found [here](#).